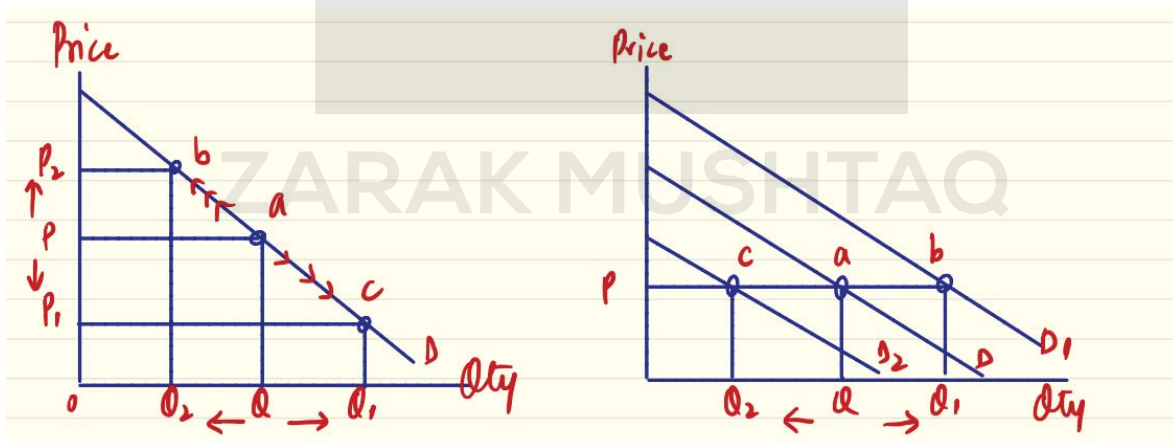


Demand & supply

Demand is a very useful tool in economics and when used in conjunction with supply can be used to predict and analyze market equilibrium. From the perspective of a producer (a business, firm, seller or supplier), demand represents the potential for sales of your products. From the perspective of a consumer (an individual, customer, purchaser or buyer), demand represents your desire and ability to consumer products or services.

Demand represents the maximum quantity of a particular good that consumers are willing and able to buy during a specified time period. For a demand to be effective; the individual should have sufficient income to buy the good. The fundamental character of the concept of demand is the relationship between the price of a good and the maximum quantity that is demanded and is described as the law of demand.

THE LAW OF DEMAND states that as the price of good increases the quantity demanded contracts. And opposite is also true, as the price of good decreases the quantity demanded extends. Movement along the demand curve will only be caused by the factor price and the other factors will remain constant i.e. keeping all other factors constant (*ceteris paribus*) or the effect of a change of one variable is being considered in isolation or when studying the effect of a price rise on demand or isolate the effect of one variable on another variable.



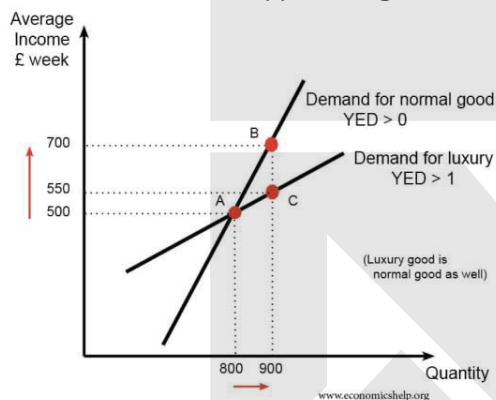
FACTORS AFFECTING DEMAND: Along a demand curve the factors, other than the price of the good, are held constant. When these factors change the demand curve shifts either outwards or inwards which means that either more or less is

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demanded at each and every price. This is called as an increase or decrease in demand.

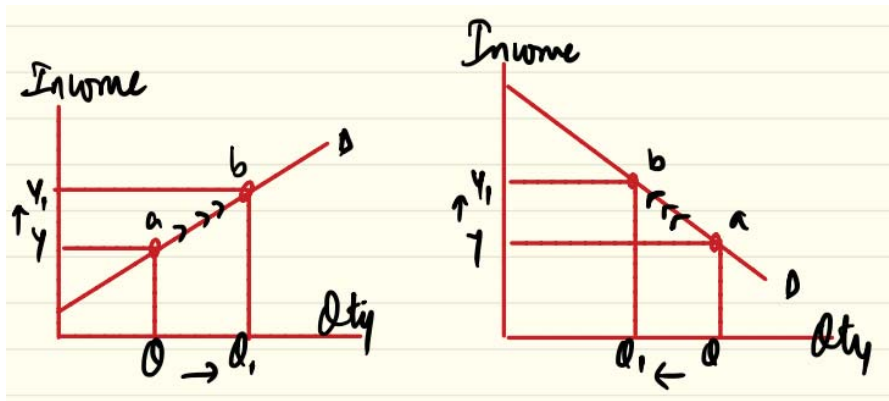
1. **THE CONSUMERS'S INCOME:** The effect that income has on the amount of a product that consumers are willing and able to buy depends on the type of good. For most goods, there is a positive (direct) relationship between a consumer's income and the amount of the good that one is willing and able to buy. In other words, for these goods when income raises the demand for the product will increase & when income falls, the demand for the product will decrease. These types of goods are called normal goods.



However, for some goods the effect of a change in income is the reverse. For example, think about a low quality (high fat-content) ground beef. You might buy this while you are a student, because it is inexpensive relative to other types of meat. But if your income increases enough, you might decide to stop buying this type of meat and instead buy leaner cuts of ground beef, or even give up ground beef entirely in favor of beef tenderloin. If this were the case (that as your income went up, you were willing to buy less high-fat ground beef), there would be an inverse relationship between your income and your demand for this type of meat. We call this type of good an inferior good. There are two important things to keep in mind about inferior goods. They are not necessarily low quality goods. The term inferior (as we use it in economics) just means that there is an inverse relationship between one's income and the demand for that good. Also, whether a good is normal or inferior may be different from person to person.

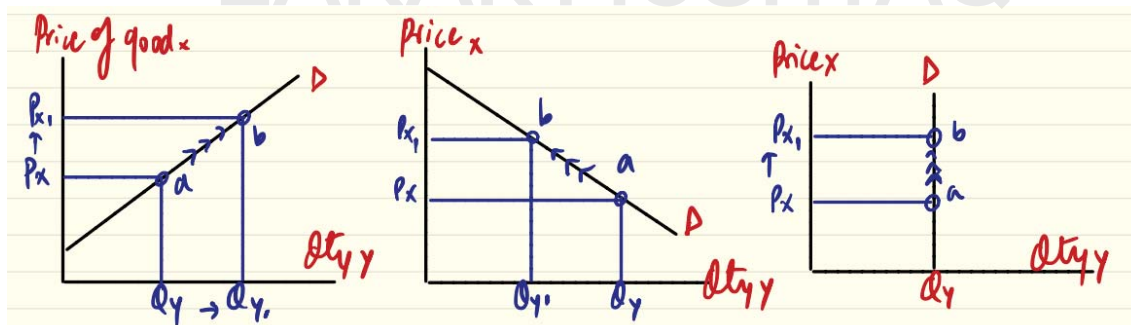
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2. **THE PRICE OF RELATED GOODS:** Two goods that are typically consumed together are called complements. If the price of car goes up, the law of demand tells us that we are willing or able to buy fewer cars and we buy less quantity of car the consumption of petrol will also decrease leading to fall in the demand. We can summarize this by saying that when two goods are complements, there is an inverse relationship between the price of one good and the demand for the other good.

On the other hand, some goods are considered to be substitutes for one another: we don't consume both of them together, but instead choose one or the other. For example, for some people coke and Pepsi are substitutes. If the price of coke increases, this may make Pepsi relatively more attractive. The law of demand tells us that fewer people will buy coke; some of these people may decide to switch to Pepsi instead, therefore increasing the amount of Pepsi that people are willing and able to buy. There is a positive relationship between the price of one good and the demand for the other good.



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3. **THE TASTE AND PREFERENCES OF CONSUMERS:** There are all kinds of things that can change one's tastes or preferences that cause people to want to buy more or less of a product. If a celebrity endorses a new product, this may increase the demand for a product. On the other hand, if a new health study comes out saying is bad for your health; this may decrease the demand for the product. Another example is that a person may have a higher demand for an umbrella on a rainy day than on a sunny day. For e.g. in the last ten years email has increasingly been used in preference to postal services; therefore postal services demand decreases while demand for email increases.
4. **CHANGES IN POPULATION:** An increase in the size of the total population leads to an increase in the demand for many goods and services causing their demand curves shift outwards. In developing countries the birth rates are very high causing the young population as a larger proportion of the total population and hence the demand for the goods and services by the young population increases.
5. **ADVERTISING:** A successful advertising campaign persuades people to buy goods and services which they originally did not need or want. Many firms spend heavily on advertising to increase their revenue and profits. For e.g. the government announces that the ingredients used in the drink can be harmful - this will likely to decrease the demand.

INDIVIDUAL AND MARKET DEMAND: Individual demand is the quantity which an individual consumer is willing to buy at a particular price. On the other hand market demand is the total quantity if a particular commodity which all the consumers are willing to buy at a particular price, so the market demand is the horizontal summation/combining of the individual demand curves or the quantity that each consumer would be willing and able to buy at each price.

PRICE ELASTICITY OF DEMAND

It measures the responsiveness to the percentage change in quantity demand to the percentage change in price of a good i.e. how the consumer respond, by making changes in the quantities which they are willing to buy, as a result of changes in the

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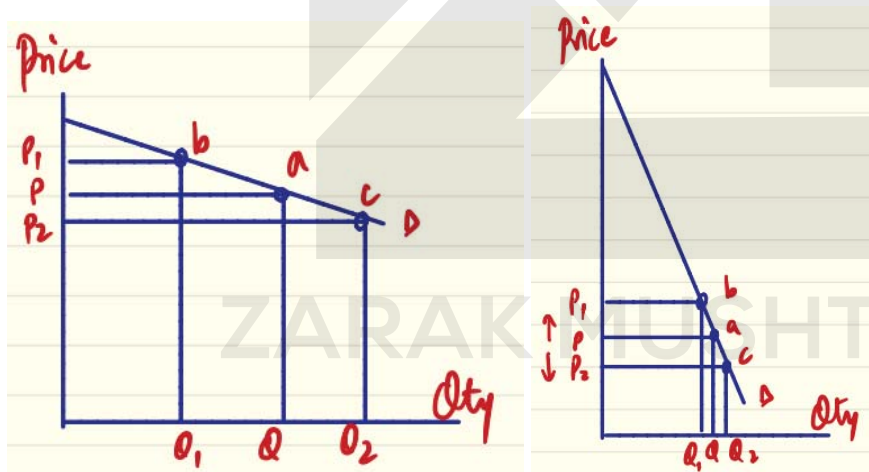
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price of a commodity. The formula for calculating the co-efficient of elasticity of demand is:

$$\text{PED} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price.}}$$

Since changes in price and quantity nearly always move in opposite directions, economists usually do not bother to put in the minus sign. We are concerned with the co-efficient of elasticity of demand. E.g. if the value of PED is -2.5 then the minus sign shows the direction of change and 2.5 is the magnitude of change which means that a change in price will bring a 2.5 times bigger change in quantity demanded; so minus sign is ignored when calculating or explaining PED.

If the percentage change in the quantity demanded is greater than the percentage change in price, the demand is said to be elastic and the value of PED will be greater than one. On the other hand, if the percentage change in quantity demanded is less than the percentage change in price, the demand is said to be inelastic and the value of PED will be less than one.



CALCULATION OF PED:

$$\% \text{change} = (\text{new} - \text{old}) / \text{old} \times 100$$

$$\text{Percentage change in price: } (P_2 - P_1) / P_1 \times 100$$

$$\text{Percentage change in quantity: } (Q_2 - Q_1) / Q_1 \times 100$$

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WHAT DETERMINES OR INFLUENCES PED?

1. **The number of close substitutes for a good / uniqueness of the product:** The closer substitutes in the market, the more elastic is the demand for a product because consumers can more easily switch their demand if the price of one product changes relative to others in the market. The huge range of package holiday tours and destinations make this a highly competitive market in terms of pricing - many holiday makers are price sensitive.
2. **The degree of necessity or whether the good is a luxury:** Goods and services deemed by consumers to be necessities tend to have an inelastic demand whereas luxuries will tend to have more elastic demand because consumers can make without luxuries when their budgets are stretched i.e. in an economic recession we can cut back on discretionary items of spending.
3. **The % of a consumer's income allocated to spending on the good:** Goods and services that take up a high proportion of a household's income will tend to have a more elastic demand than products where large price changes makes little or no difference to someone's ability to purchase the product.
4. **The time period allowed following a price change:** Demand tends to be more price elastic the longer we allow consumers to respond to a price change by varying their purchasing decision. In the short-run, the demand may be inelastic, because it takes times for consumers both to notice and then to respond to price fluctuations.
5. **Whether the good is subject to habitual consumption:** When this occurs, the consumer becomes much sensitive to the price of the good in question. Clearly, examples such as cigarettes and alcohol and other drugs can come into this category; so if the good is habit forming, this is likely to make good inelastic.

EXTERME CASES OF PED:

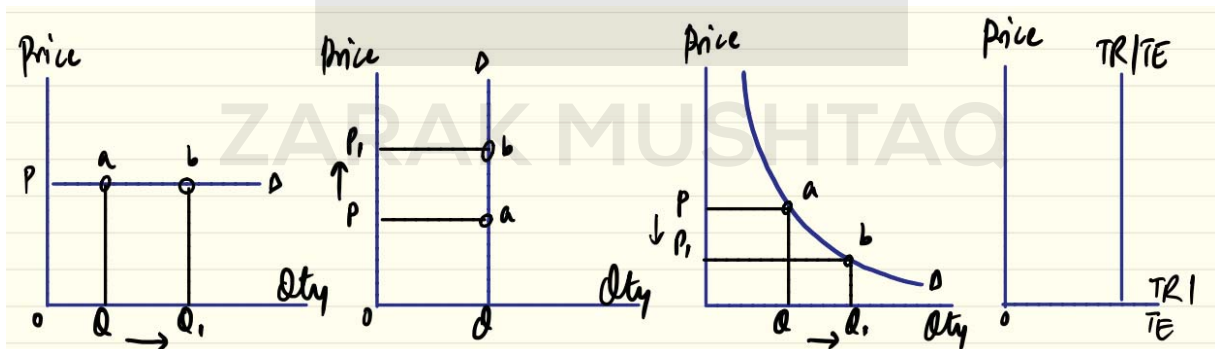
1. **PERFECTLY ELASTIC DEMAND:** In this situation a minute or fractional change in price brings very large or infinite changes in quantity demanded and the coefficient of PED is equal to infinity. In perfectly competitive market where there are a large number of producers selling a particular homogenous commodity at a particular market price faces a perfectly elastic

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(horizontal) demand curve. If one of the producers increase the price of his product a little bit the demand for his product will fall to zero i.e. no consumer will buy from him and if he decreases the price a little bit then the demand for his product will increase infinitely. Percentage change in quantity demand is equal to the percentage change in total revenue.

2. **PERFECTLY INELASTIC DEMAND**: In this situation a change in price brings no change in quantity demanded and the value of PED is equal to zero. There are very few examples of perfectly inelastic demand where a consumer buys the similar quantity at any price e.g. life saving drugs, blood. In this situation a change in the price of the good brings an equal percentage change in the revenue of the producer or the expenditure of the consumer. When all economics teachers are dead and only ZM lives - perfectly inelastic demand
3. **UNITARY ELASTIC DEMAND**: In this situation a change in price brings an equal percentage change in quantity demanded and the value of PED is one. In this case the total revenue of the producer or the expenditure of the consumer remains unchanged as a result of changes in the price of the good. There is no particular good which always show unitary elastic demand, it happens when a consumer fixes his expenditures on the purchase of a good. The shape of a unitary elastic demand curve is called rectangular hyperbola which shows that the area of the percentage change in price and the area of the percentage change in quantity demanded are always equal.



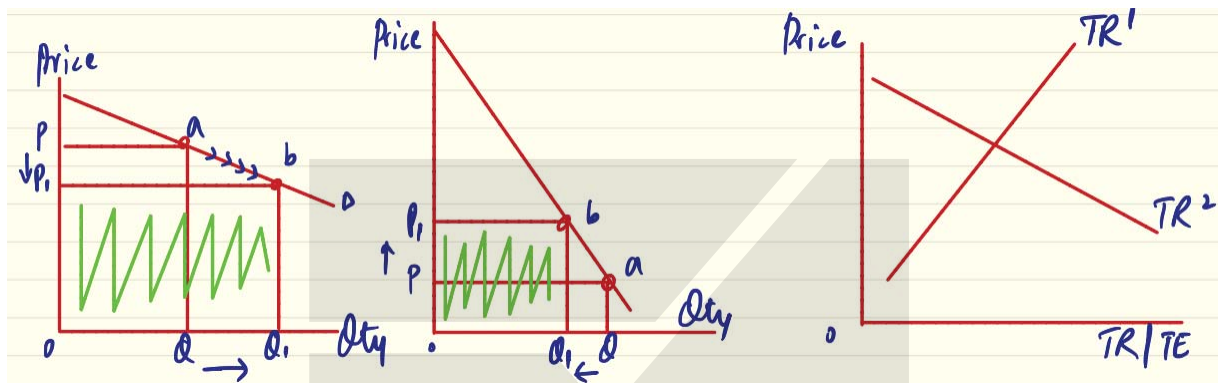
PED & Total revenue (total expenditure):

The revenue (income) of the producer depends very much on the elasticity of demand for the good which he is selling. With the knowledge of PED for his good he can make appropriate changes in the price of the good to increase his total

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revenue. In case of product being elastic, a change in price brings a larger proportionate change in quantity demanded, so producer should decrease the price to increase total revenue of the firm; while in case of inelastic product, a change in price brings a lesser proportionate change in quantity demanded, so producer can increase price to increase its revenue.



Usefulness of PED to a government: If a government is aware about the elasticity of demand then it can set its objective and achieve it accordingly. In case of elastic demand, when a government imposes indirect tax, it can reduce the consumption while in case of demand being inelastic, imposing indirect tax will result in tax revenue.

On the other hand, if the government wants to encourage firms and increase sales revenue, it will provide subsidy to the firm with demand being elastic and if the government wants to reduce the expenditure on the individual then providing subsidy with inelastic demand will achieve its objective.

An objective of some firms is to make the PED for their product more inelastic or elastic

An objective of some firms is to make the PED for their product more price inelastic in order to increase revenue. This can be achieved in various ways such as:

- Firms use **persuasive advertising** to try to influence (persuade) consumers to buy a product. Persuasive advertising highlights the benefits of the product compared to its substitutes. For example, this type of advertising may emphasize the health benefits of a product or a well-known celebrity may promote (endorse) the product. It results in a shift to the right of the

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demand curve for the product, with increased sales. There is no change in price.

- A firm may create a **brand image (branding)** for its products in order to make the products appear superior when compared to similar products from competitors. Branding is often used by multinational companies to persuade consumers to buy their brand rather than similar supermarket own brands.
- A firm **takes over or merges** with a competitor to increase market share and therefore control over a market.
- A firm creates a **monopoly product** where a firm is the only producer of the product which is protected by a patent or regulations.

Making a product more **price elastic** is more difficult. It can be done if a product is one of many products in a market where demand is particularly price sensitive. Retailers often use special sales promotions to increase sales by reducing the price of the product with the intention that consumers continue to purchase the product when the promotion has ended. Such a policy is risky since competitors are likely to make the same price reduction, in which case all producers will probably lose out.

CROSS ELASTICITY OF DEMAND (XED): It measures the responsiveness to the percentage change in quantity demand for good A following a percentage change in the price of good B. We are mainly concerned here with the effect that changes in relative prices within a market have on the pattern of demand.

$$\text{XED} = \frac{\text{Percentage change in quantity demanded of good A}}{\text{Percentage change in price of good B}}$$

Substitutes: With substitute goods such as brands of cereal or washing powder, an increase in the price of one good will lead to an increase in demand for the rival product. Cross price elasticity for two substitutes will be positive.

Complements: With goods that are in complementary demand, such as the demand for DVD players and DVD videos, when there is a fall in the price of DVD players we expect to see more DVD players bought, leading to an expansion in market demand for DVD videos. The XED for two complements is negative.

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For example with two close substitutes, the XED will be strongly positive. Likewise when there is a strong complementary relationship between two products, the XED will be highly negative. Unrelated products have a zero XED.

How can firms make use of XED to increase total revenue?

Pricing strategies for substitutes: If a competitor cuts the price of a rival product, firms use estimates of XED to predict the effect on the quantity demanded and the total revenue of their own product.

Pricing strategies for complementary goods: For e.g. a popcorn, soft drinks and cinema tickets have a high negative value for XED - they are strong complements. Popcorn has high markup i.e. popcorn costs pennies to make but sells for more than a pound. If firms have a reliable estimate for XED they can estimate the effect, say, of two-for-one cinema ticket offer on the demand for popcorn. The additional profit from extra popcorn sales may more than compensate for the lower cost of entry in to the cinema.

Advertising and marketing: In highly competitive markets where brand names carry substantial value, many businesses spend high amounts of money every year on persuasive advertising and marketing. There are many aims behind this, including shifting out the demand curve for a product (or product range) and also building consumer loyalty to a brand. When consumers become habitual purchasers of a product, the XED against rival will decrease. The result is the firms may be able to charge a higher price, increase their total revenue and turn consumer surplus into higher profits.

Income elasticity of demand

It measures the responsiveness to the percentage change in quantity demanded due to percentage change in consumer income.

$$YED = \frac{\text{Percentage change in Quantity demanded}}{\text{Percentage change in income.}}$$

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Values of income elasticity tell us something about the nature of a product and how it is perceived by consumers, it also affects the extent to which it changes in economic growth. It affects the level and pattern of demand for different goods and services.

YED is used to classify goods in relation to changes in the incomes of consumers. This classification is based on the size (in numerical terms) and the sign (positive or negative) of the YED.

Four types of good can be identified using YED:

- A **normal good** is one where the quantity demanded increases as income increases. Most products are like this. In China and in India the demand for chicken meat has increased as incomes have increased for many households. In emerging economies especially, the demand for cars increases as incomes increase as a result of economic growth. Elsewhere, even in very poor countries, a smartphone is now regarded as a normal good. For normal goods, the YED is positive and expected to be between 0 and 1.
- An **inferior good** is one where the quantity demanded decreases as income increases or increases as income falls. In the case of a rise in income, consumers use their increased incomes to buy better quality goods to replace inferior substitutes. Typical examples of inferior goods are poor quality rice and packet noodles. The YED is negative; its size indicates the strength of the relationship between a change in income and the quantity demanded.
- A **necessity good** is a type of normal good for which the quantity demanded is unlikely to change when income changes. Staple foodstuffs such as rice, flour and pulses that are regularly consumed are typical. The YED is positive but is likely to be close to zero. A low value for the YED indicates that there is a limit to the quantity of these goods that households purchase even when the change in income might seem to be substantial.
- A **superior or luxury good** has a positive YED that is greater than 1 and is a normal good where the quantity demanded is responsive to changes in income. The higher the size of the YED, the greater the change in quantity demanded. Examples of superior goods are designer clothes and jewelry or

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the latest electronic devices. In low-income economies, a motor cycle can be a superior good compared to using the bus or a bicycle for example.

Factors affecting YED: The major factor determining the income elasticity of demand (YED) of a product is the extent to which it is regarded as either a necessity or a luxury. Necessities are goods or services which are regarded by individuals as essential and which they will purchase irrespective of changes in their income. Examples might include food, utilities such as gas and water, vital medicines such as insulin for diabetics, as well as certain addictive or habit forming products such as cigarettes. Luxuries, on the other hand, tend to be expensive high quality products whose demand increases disproportionately as an individual's income increases and they have more money available for discretionary expenditure. Examples might include expensive perfumes, designer clothes and high quality accessories such as Hermes Birkin handbags. Luxuries will generally have substitutes, but these are regarded as less desirable.

It is important to acknowledge, however, that, as indicated earlier, the distinction between necessities and luxuries is not an absolute one as what one individual might regard as a necessity, another may regard as a luxury, e.g. the latest mobile phone. Even within product categories such as food individual products may be viewed differently by individuals.

At this point it might be useful to review our understanding of normal and inferior goods. Inferior goods generally have superior substitutes and the demand for them varies inversely with changes in an individual's income; increasing when incomes fall and decreasing when incomes rise.

Normal goods are those for which the demand varies directly with changes in an individual's income; increasing when incomes rise and vice versa.

Necessities are either inferior goods whose demand falls as income rises and as such have a negative income elasticity of demand ($YED < 0$) or normal goods in that the demand for them rises or falls by a smaller percentage than the percentage change in income. In this case the demand is income inelastic ($YED < 1$). For example, an individual or household on a very low income may satisfy its need for food by purchasing canned processed meat and vegetables and supermarket own

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brands, but as their income rises they are able to afford more nutritious fresh meat and vegetables and branded goods. Hence the demand for tinned food and supermarket own brands declines and the demand for fresh food and branded goods increases, but by a smaller proportion than the increase in income. Necessities occupy a lower proportion of household income as incomes rise. For luxuries the demand increases rapidly as income changes which means that they take up a higher proportion of household income as incomes increase. The percentage increase in demand for luxuries is greater than the percentage change in income. Demand is income elastic ($YEO > 1$).

It is also important to note that the income elasticity of demand can be affected by how broadly or narrowly a product or product category is defined. Generally, the broader the definition of a product, the more inelastic the demand is likely to be. For example, overall the income elasticity of demand for food is inelastic, but within this product grouping there are items which have an elastic demand with respect to income such as asparagus or exotic cheeses.

It is also important to remember that throughout this analysis we are assuming that the only variable affecting the demand for the product that is changing is household income and that all other determinants of demand are held constant.

SUPPLY

Supply represents the minimum quantity of a particular good that producers are willing and able to sell during a specified time period.

LAW OF SUPPLY: It states that as the price of good increases the quantity supplied extends and opposite is also true, as the price of good decreases the quantity supplied contracts. This makes sense intuitively since firms would likely be more willing to sell more of a good for more money rather than less of it.

There are three reasons why supply curves for most products are drawn as sloping upwards from left to right giving a positive relationship between the market price and the quantity supplied:

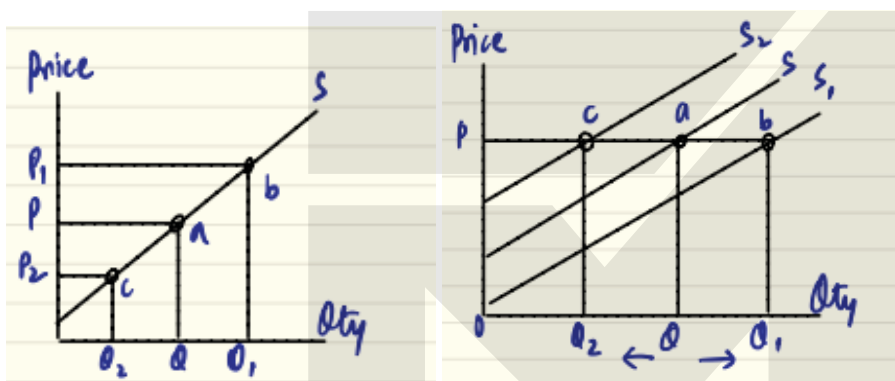
1. **THE PROFIT MOTIVE:** When the market price rises (for example after an increase in consumer demand), it becomes more profitable for businesses to

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increase their output. Higher prices send signals to firms that they can increase their profits by satisfying demand in the market.

2. **PRODUCTION AND COSTS:** When output raises, a firm's production costs may rise, therefore a higher price is needed to justify the extra output and cover these extra costs of production.
3. **NEW ENTRANTS INTO THE MARKET:** Higher prices make it more profitable for other firms to enter the market leading to an increase in supply available for consumer to buy.

**FACTORS AFFECTING SUPPLY:**

1. **The price of inputs:** In addition to the price of the product being the main factor as stated in the law of supply, the price of production inputs also plays a part. The lowest price at which a firm can sell a good without losing money is the amount of money that it costs to produce it. Producing a good or service involves taking inputs and applying a process to them to produce an output. The output is the finished good or service, and inputs are raw materials, labor, utilities and other goods. These inputs are also known as factors of production. If the price of inputs goes up, the cost of producing the good increases and therefore at each price producers need to sell their good for more money. So an increase in the price of inputs leads to a decrease in supply. Similarly, a decrease in the price of inputs leads to an increase in supply. For e.g. a rise in the wages of diamond miners will decrease the supply as the cost of production will be increased. For e.g. rise in building cost will decrease the supply. For e.g. a fall in the price of paper will increase the supply of a firm who publishes and sells books.

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2. **The current state of technology:** Production technology is involved in the process part. Increases in the level of production technology can make that process more efficient with the use of technology, the operation becomes more efficient. For example, imagine that you run a basic T-shirt screen printing business out of your home. Now let's say you decide to invest in a workshop installed with the latest production technology. With this use of technology, the operation becomes more efficient and you are able to increase the supply of T-Shirts. If you decide to expand even further, some added technological improvements might be warranted. This further increases your ability to supply t-shirts since it reduces labor costs. By automating the process, reliance upon laboratories is lessened and those resources are released for utilization elsewhere.
3. **Taxation and subsidies:** Taxes imposed on goods and services are called indirect taxes e.g. sales tax. The imposition of indirect tax will bring about changes in supply. A tax on a commodity may be regarded as an increase in the costs of production. The producer then reacts by raising the price of the product through reduction in the supply of the goods i.e. the supply curve moves to the left. The vertical distance between two supply curves shows the per unit tax imposed by the government. For e.g. tax on computer games
A subsidy, on the other hand, is a financial help provided by the government to encourage the producer to produce more output. A subsidy acts opposite to the tax i.e. it tends to reduce the cost of production causing the supply curve to shift to the right. The vertical distance between two supply curves shows the per unit subsidy by the government.

Market supply is the result of aggregating the supply of all individual firms. The price of the product is not held constant when aggregating individual firm's supply curves to give the market supply curve.

PRICE ELASTICITY OF SUPPLY (PES):

It measures the responsiveness to the percentage change in quantity supplied to the percentage change in price i.e. how the producers behave, by making either big

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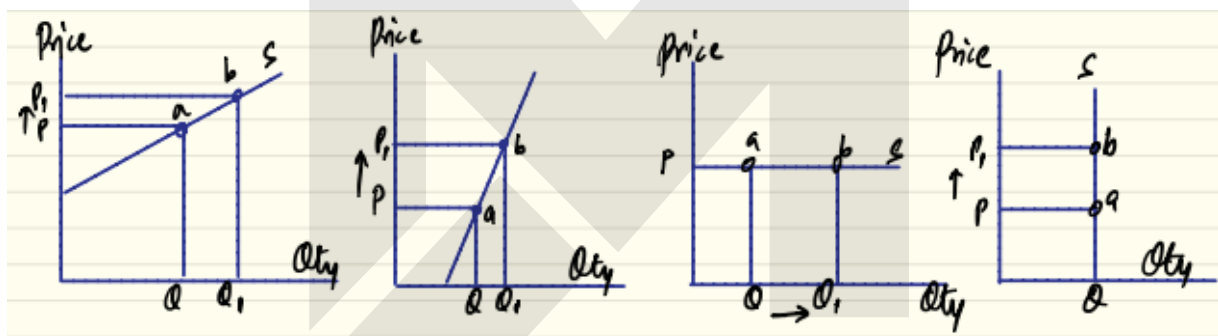
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or small changes in the quantity which they are willing to sell, in response to changes of the good.

PES = Percentage change in quantity supplied of good X

Percentage change in price of good X

If the percentage change in quantity supplied exceeds the percentage change in price, supply is said to be elastic i.e. the producer is responsive to the change in price. In this case the value of PES will be greater than 1. On the other hand, the percentage change in quantity supplied is lesser than the percentage change in price, supply is said to be inelastic. The value in this case will be less than 1. The value of price elasticity is positive because an increase in price is likely to increase the quantity supplied to the market and vice versa.



Factors or determinants of PES:

1. **Spare production capacity**: If there is plenty of spare capacity, a business should be able to increase its output without a rise in costs and therefore supply will be elastic in response to a change in demand. The supply of goods and services is often most elastic in a recession when there is plenty of spare labor and capital resources available to setup output as the economy recovers.
2. **Stocks of finished products and components**: If stocks of raw material and finished products are at high level then a firm is able to respond to a change in demand quickly by supplying these stocks on to the market, supply will be elastic. Conversely when stocks are low, dwindling supplies force prices

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higher and unless stocks can be replenished, supply will be inelastic in response to a change in demand.

3. **The nature of the commodity:** In the case of agricultural commodities, supply is inelastic in the short run because agricultural production depends on the harvesting period. Some agricultural commodities have inelastic demand for many years e.g. coffee plant takes several years to produce coffee beans. Thus in the short run an increase in the demand for coffee will lead to a sharp increase in its price. The supplies of products beef and milk will also be inelastic because it takes a considerable time to increase the sizes of herds of beef and dairy cattle.
4. **The ease and cost of factor substitution:** If both capital and labor resources are occupationally mobile then the elasticity of supply for a product is higher than if capital and labor cannot easily and quickly be switched.
5. **Time period:** Elasticity of supply increases with time as producers have longer time to adjust to changes in demand. Supply of most of the commodities is fixed in a particular moment in time where the PES is zero and is referred to as momentary - run. In the short run supply can be increased as a result of an increase in the price but it is usually inelastic i.e. PES will be less than 1. In the long run supply tends to be more elastic i.e. the value of PES will be greater than 1. For e.g. a decrease in the time that it takes to produce the products would increase the PES of a firms' products.

Joint supply is an economic term referring to a product or process that can yield two or more outputs. Common examples occur within the livestock industry: cows can be utilized for milk, beef, and hide. Sheep can be utilized for meat, milk products, wool, and sheepskin. If the supply of cows increases, so will the joint supply of dairy and beef products. An increase in demand for a product will increase the supply of jointly supplied good.

EXTREME CASES:

1. **Perfectly elastic supply:** It is a situation where a minute change in the price of a commodity leads to infinite changes in the quantity supplied and the

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value of PES is infinite. In this case the supply curve will be a horizontal line which shows that the producer must sell any quantity that he wants at a particular price.

2. **Perfectly inelastic supply:** It is a situation in which a proportionate change in price brings no change in quantity supplied i.e. the supply is fixed and cannot be changed in response to changes in price. In this case the value of PES will be zero and the supply curve will be a vertical line. There can be many examples of perfectly inelastic supply e.g. number of car parking spaces in a parking lot, number of seats in a concert hall or stadium, antiques. In all these cases the supply is fixed and it cannot be increased in the short run.

3. **Unitary elastic supply:** In this situation a change in price brings an equal proportionate change in quantity supplied and the value of PES will be 1 throughout the supply curve. In this case the supply curve will be a straight line starting from the origin.



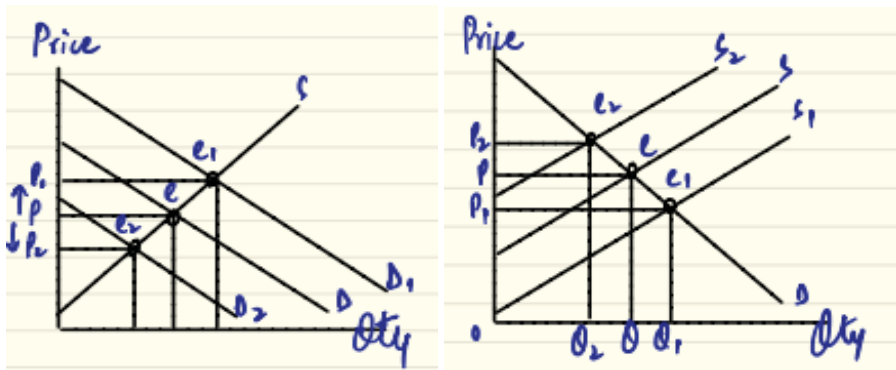
MARKET EQUILIBRIUM

Equilibrium is a state of rest or balance due to the equal action of opposing forces. Equilibrium price is the price towards which the market forces of demand and supply are balanced. When a market is in equilibrium, there is no excess supply or demand. Equilibrium quantity is the amount bought and sold at the equilibrium price. The change in equilibrium position is shown below in different situations i.e. increase in demand, decrease in demand, increase in supply and decrease in supply.

Price mechanism is where prices are determined by the forces of demand and supply where prices provides signal to the producer to allocate the resources in profit maximizing manner.

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Derived demand applies where something is required because it is needed for the production of other goods and services. A typical example is labour, a factor of production. A hotel business demands labour in order to provide accommodation and food for its visitors. If there is an increase in demand for the hotel's services, the demand for hotel workers will also increase. Therefore, the demand for hotel workers is derived from the demand for the hotel's services.

Joint supply

Joint supply occurs when two goods are produced together but for different purposes. An example is the case of soya bean processing in the US where the crop is first processed for human consumption and the remainder is made into animal feed. Figure 10.14 shows how an increase in demand for processed soya for humans leads to an increase in supply and a fall in price of the animal feed.

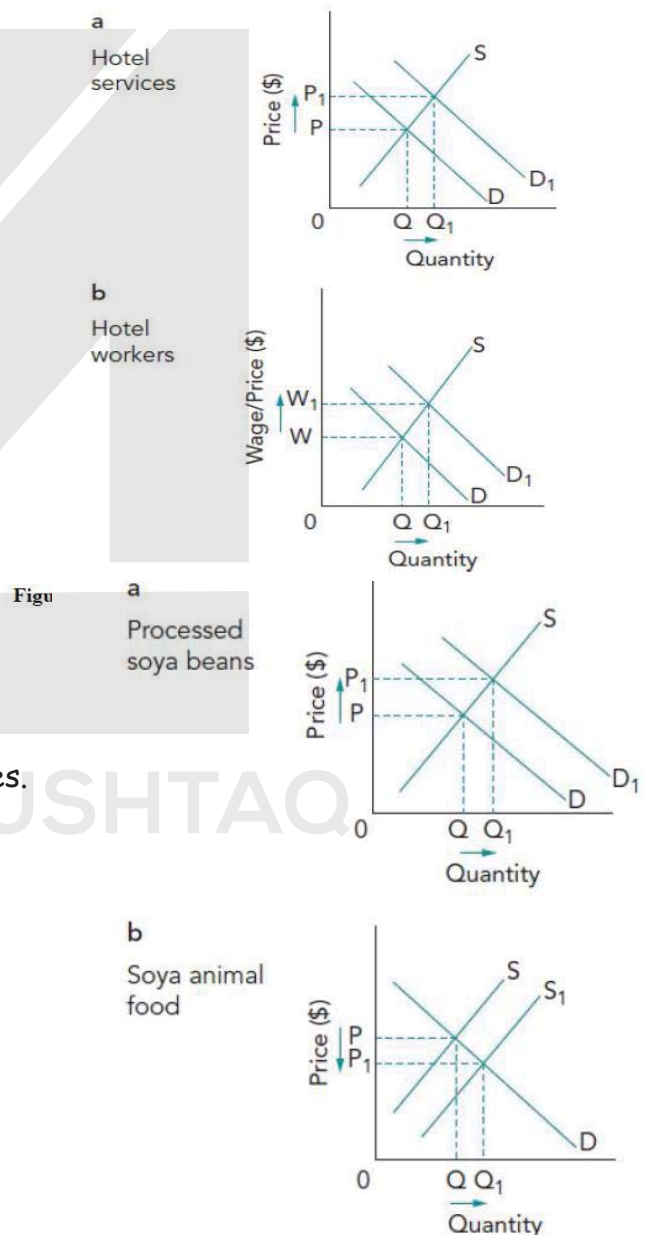


Figure 10.14: Joint supply

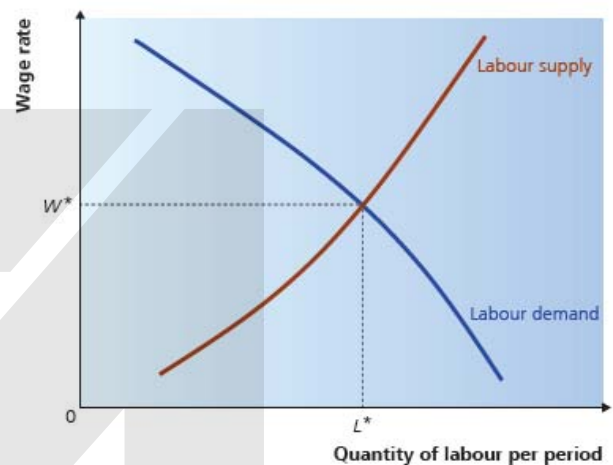
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Derived demand: There are situations in which the demand for a good or service is indirect. This is best seen through some examples.

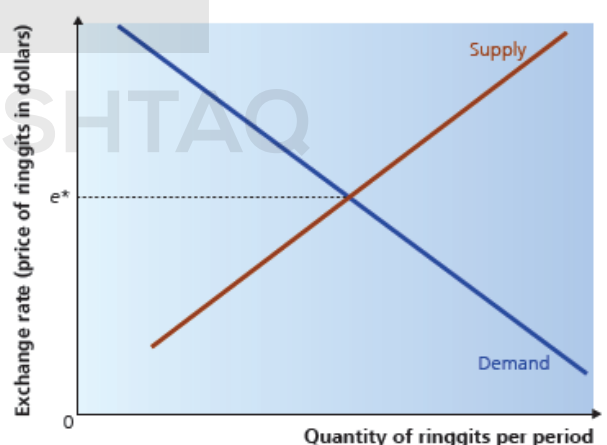
The labour market: Within the economy, firms demand labour and employees supply labour – so can we use demand and supply to analyze the market for a factor of production?

From the firms' point of view, the demand for labour is a derived demand. In other words, firms want labour not for its own sake, but for the output that it produces. When the price of labour is low, firms will tend to demand more of it than when the price of labour is high. The wage rate can be regarded as this price of labour. On the employee side, it is argued that more people tend to offer themselves for work when the wage is relatively high. On this basis, the demand for labour curve is expected to be downward sloping and the supply of labour curve upward sloping. As usual, the equilibrium in a free market will be at the intersection of demand and supply, so firms will hire L^* labour at a wage rate of W^* .



▲ Figure 4.9 A labour market

The foreign exchange market: If you were to take a holiday in the USA, you would need to buy dollars. Equally, when American tourists travel to visit Malaysia or Mauritius, they need to buy ringgits or rupees. If there is buying going on, then there must be a market. So here is another sort of market to be considered. The exchange rate is the price at which two currencies exchange, and it can be analyzed using demand and supply. Consider the market for ringgits, and focus on the exchange rate between ringgits and dollars.



▲ Figure 4.10 The market for ringgits

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Think first about what gives rise to a demand for ringgits. It is not just Americans visiting Malaysia who need dollars to spend on holiday: anyone holding dollars who wants to buy Malaysian goods needs ringgits in order to pay for them. So the demand for ringgits comes from people outside Malaysia who want to buy Malaysian goods or services - or, indeed, assets. When the exchange rate for ringgits in terms of dollars is high, potential buyers of American goods get relatively few ringgits per dollar, so the demand will be relatively low, whereas if the dollar per ringgit rate is relatively low, they get more for their money. Hence the demand curve is expected to be downward sloping.

Foreign exchange is a derived demand, in the sense that people want ringgits not for their own sake, but for the goods or services that they can buy. One way of viewing the exchange rate is as a means by which to learn about the international competitiveness of exports from the USA. When the exchange rate (dollars per ringgit) is high, Malaysian goods are less competitive in the USA, *ceteris paribus*. Notice the *ceteris paribus* assumption here. This is important because the exchange rate is not the only determinant of the competitiveness of Malaysian goods: this also depends on the relative price levels in Malaysia and the USA.

How about the supply of ringgits? Ringgits are supplied by Malaysians wanting dollars to buy goods or services from the USA. From this point of view, when the dollar/ ringgit rate is high, Malaysian residents get more dollars for their ringgits and therefore will tend to supply more ringgits.

If the exchange market is in equilibrium, the exchange rate will be at e^* , where the demand for ringgits is matched by the supply

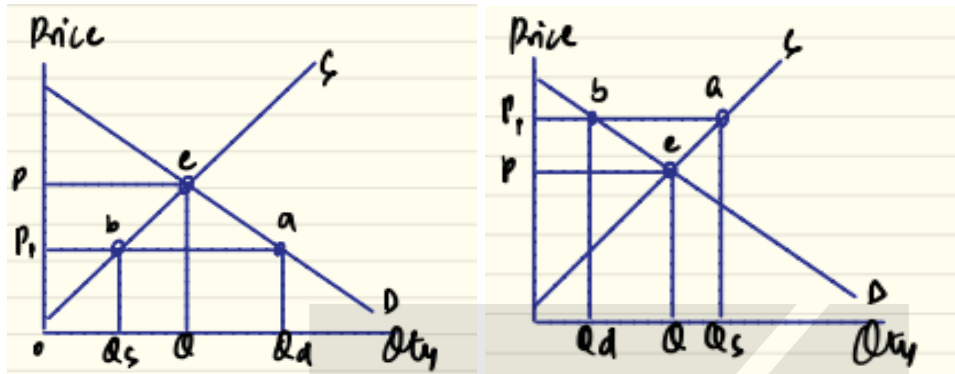
MARKET SURPLUSES & SHORATAGE

A market surplus occurs where there is excess supply that is quantity supplied is greater than quantity demanded. In this situation some producers won't be able to sell all their goods. This will induce them to lower their price to make their product more appealing. In order to stay competitive many firms will lower their prices thus lowering the market price of a product. In response to the lower price, consumers will increase their quantity demand, moving the market toward an equilibrium price

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and quantity. In this situation, excess supply has exerted downward pressure on the price of the product.



A market shortage occurs when there is excess demand - that is quantity demanded is greater than quantity supplied. Consumers won't be able to buy as much good as they would like. Producers will raise both the price of their product and the quantity they are willing to supply. The increase in price will be too much for some consumers and they will no longer demand the product. Eventually equilibrium will be achieved.

FUNCTIONS OF THE PRICE MECHANISM

The invisible hand - the workings of the price mechanism: The price mechanism is a term used to describe the means by which the many millions of decisions taken each day by consumers and businesses interact to determine the allocation of scarce resources between competing uses. This is the essence of economics. The price mechanism plays three important functions in any market based economic system:

The signaling functions: Firstly, prices perform a signaling function. This means that market prices adjust to demonstrate where resources are required, and where they are not. Prices rise and fall to reflect shortages and surpluses. So, for example, if market prices are rising because of stronger demand from consumers, this is a signal to suppliers to expand their production to meet the higher demand.

The transmission of preferences: Through the signaling function, consumers through their expression and preferences send important information to producers about the changing nature of our needs and wants. When demand is strong, high

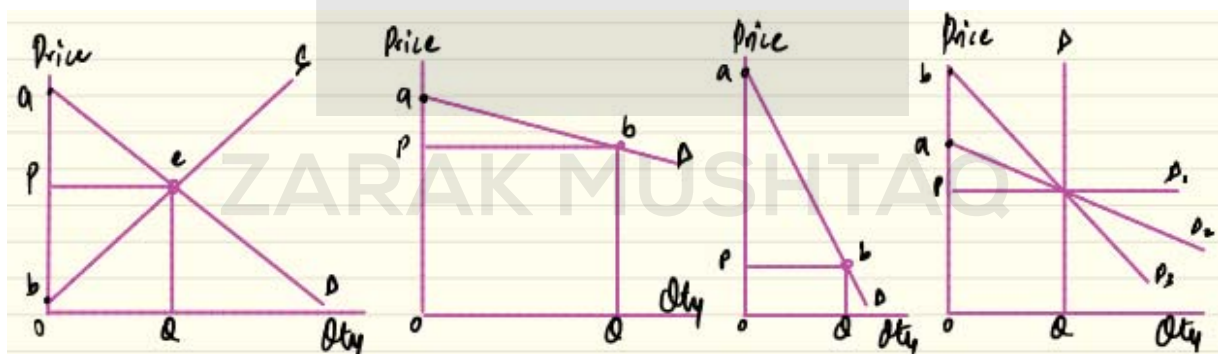
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market prices act as an incentive to raise output (production) because the suppliers stand to make a higher profit. When demand is weak, then market supply contracts.

The rationing function: Prices serve to ration scarce resources when demand in a market outstrips supply. When there is a shortage of a product; the price is bid up - leaving only those with sufficient willingness and ability to pay with the effective demand necessary to purchase the product. The market price acts a rationing device to equate demand and supply.

CONSUMER SURPLUS: Consumer surplus is the difference between the price which the consumer is willing to pay and the price which he actually pays. The price which a consumer is willing to pay for a particular unit of a commodity and the price which he pays for that unit is called the marginal consumer surplus' which is the additional or surplus of benefit over and above the price of that unit. For example if a consumer is willing to pay \$10 for a pair of shoes but the market price is \$8 per pair, his consumer surplus will be \$2. As consumer buys more units of a commodity, he is willing to pay lesser and lesser price, so his marginal consumer surplus tends to fall until it reaches zero. At this point this total consumer surplus will be maximized. The total consumer surplus is calculated by adding all the marginal consumer surpluses.

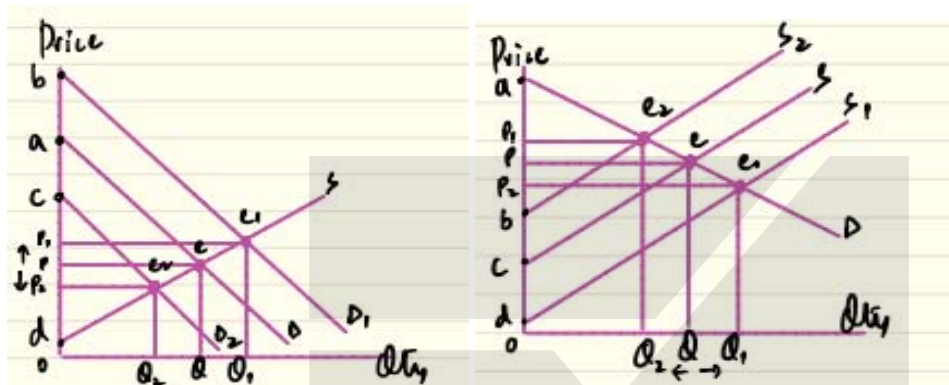


CONSUMER SURPLUS AND PRICE ELASTICITY OF DEMAND: When demand for a good or service is perfectly elastic, consumer surplus is zero because the price that people pay matches the price they are willing to pay. When demand is perfectly inelastic, consumer surplus is infinite. Demand is totally invariant to a price change. When the demand is inelastic, there is a greater potential consumer

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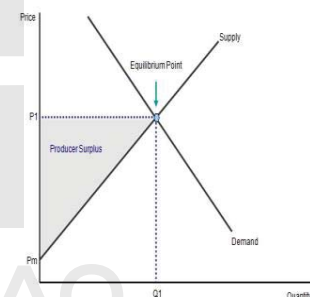
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surplus because there are some buyers willing to pay a higher price to continue the product. The area below the demand curve and above the price level will be the consumer surplus. When the demand is inelastic the consumer surplus will be higher as compared to in the case of elastic demand the consumer surplus will be less. Consumer surplus will be zero if demand and supply are perfectly elastic.



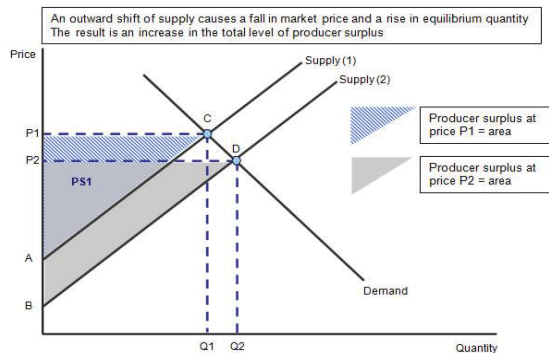
Changes in demand and supply can affect consumer surplus. For e.g. an increase in supply will reduce equilibrium price, keeping the maximum price willing to pay remains unchanged, and this increases consumer surplus.

Producer surplus: Producer surplus is a measure of **producer welfare**. It is measured as the difference between what producers are willing and able to supply a good for and the price they actually receive. The level of producer surplus is shown by the area above the supply curve and below the market price and is illustrated in the diagram below P_m is the minimum price that this producer requires to supply the product to the market. As the price rises, there is a great incentive to supply - production will expand as a business moves up their supply curve.



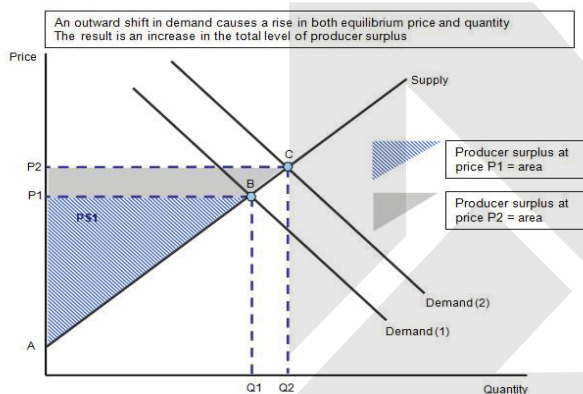
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We first consider the effects of a change in market supply - for example caused by an improvement in production technology or a fall in the cost of raw materials and components used in the production of a good or service

We now consider the effects on producer surplus of a rise in market demand

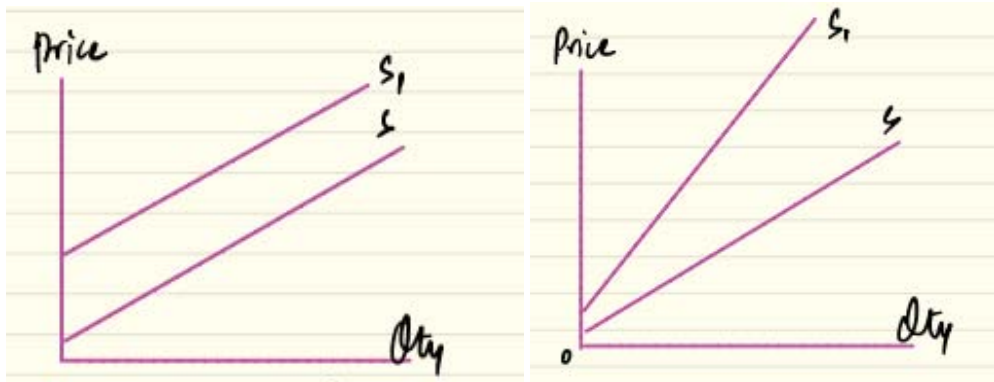


GOVERNMENT INTERVENTION IN MARKETS: Governments sometime seek to alter market outcomes. They can influence the prices at which goods are sold by taxing or subsidizing either their production or their consumption. For example if government wants to discourage the production and consumption of alcohol, it can impose heavy indirect taxes on the producers. On the other hand if the government wants to encourage the production of wheat, it will provide subsidies to the farmers.

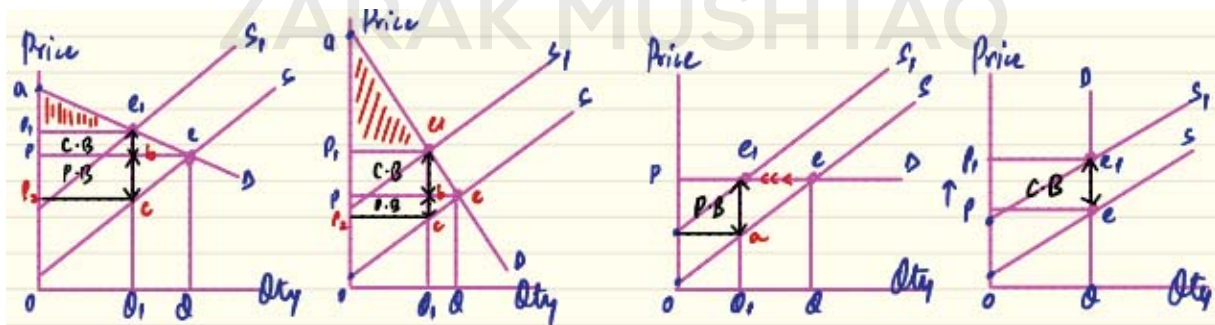
INDIRECT TAXES: An indirect tax can be imposed in two ways i.e. (1) specific tax; a fixed amount of tax per unit of commodity e.g. \$1 per gallon of petrol (2) ad-valorem; some percentage of the price of the commodity e.g. 15% of the price of pizza. Specific tax is the vertical distance between two supply curves.

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When an indirect tax is imposed on the good, it leads to an increase in the cost of production and is shown by a leftward shift in the supply curve. In the case of specific tax it will be a parallel shift because the tax amount is constant at all prices and in the case of ad-valorem tax, the new supply curve will diverge away from the original supply curve at higher and higher prices. The vertical distance between the two supply curves is the per unit tax. The supply curve shifts to left by the amount of tax because the price needs to increase by the amount of tax to persuade the producers to continue to produce the same quantity that he sells and increase the price but since the demand curve is downward sloping the price does not rise by the amount of tax. Thus the burden of tax is shared by the producers and consumers. The burden of tax is referred to as the incidence of tax which depends on the particular supply and demand conditions in the market. Whether the tax is borne by the producers or whether they can pass the greater part of it on to consumers in the form of a higher price depends largely on the elasticity of demand and supply.



INCIDENCE/BURDEN OF TAX & REVENUE FOR GOVERNMENT: When the demand is inelastic, the increase in price is greater as compared to the increase in

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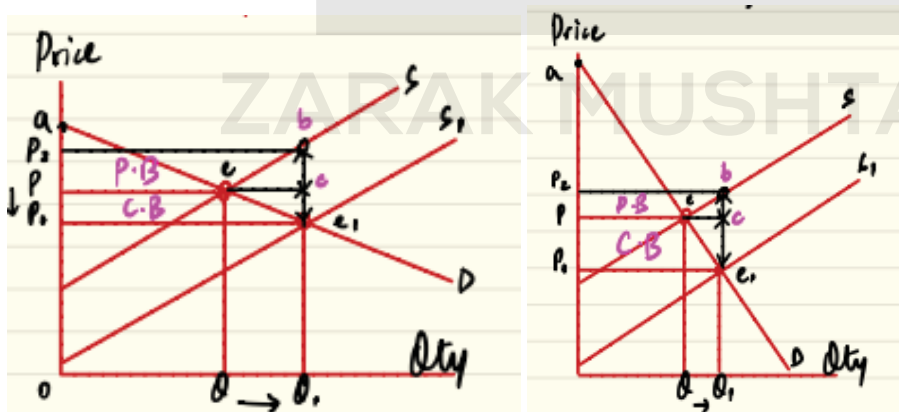
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the price in the case of elastic demand. The reason is that the producer can increase the price easily where the demand is inelastic because the consumers continue to buy more or less the same quantity of the commodity. However in the case of elastic demand the producer will not be able to increase the price because there will be a large fall in the quantity demanded so he bears the greater burden.

In general if the elasticity of demand is greater than elasticity of supply, the producers bear the greater share of the tax burden; if the elasticity of demand is less than the elasticity of supply, the consumers bear the greater share.

The two main reason for a government placing a tax on a good or service is to raise revenue or to discourage the consumption and production. If the government wishes to raise large sums of money from indirect taxation, it must concentrate heavily on those commodities for which the demand is inelastic but if the prime objective is to reduce the output and consumption of the product, it is more advantageous if demand is elastic.

Subsidy: It is a grant or a cash payment given by the government to the producers to encourage them to produce more. This reduces their cost of production and encourages them to supply more. As shown in the diagram below; an increase in supply from S_1 to S_2 will reduce the equilibrium price from P_1 to P_2 , extending the equilibrium quantity from Q_1 to Q_2 ; therefore a new equilibrium point will be achieved.



Subsidies may be provided for many reasons including:

- To keep down the market prices of essential goods

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- To encourage greater consumption of merit goods
- To contribute to a more equitable distribution of income
- To provide services that would not be provided by the free market
- To raise producers' income, especially in the case of farmers
- To provide an opportunity for exporters to sell more goods
- To reduce dependence on imports by paying subsidies to domestic producers of close substitutes.

Estimating the size of subsidy required is a further difficulty. Another problem is that subsidies are 'blanket' or lump sum payments and, unlike taxes on consumers, cannot easily be linked to incomes and the ability to pay. Consequently, in the case of food items both high-income and low-income consumers pay the same amount per unit of what they consume. It is therefore necessary to assess who benefits from a particular subsidy. This is shown by the following examples:

Staple foods like rice, bread and cooking oil are subsidized in some low and lower middle-income economies. As world food prices continue to rise, the case for subsidy is a strong one, mainly to provide relief for the lowest income groups in the economy. However, subsidized prices are paid by all income groups, many of whom can afford to pay more. There could also be shortages since demand is greater than supply. If the subsidy is paid to producers, there is no incentive for inefficient producers to improve; neither is it certain that the money received as a subsidy payment will be used for its intended purpose.

Mass transit (public transport) is heavily subsidized in all parts of the world, especially in cities and large towns. This is done to give low earners access to employment opportunities, to provide social mobility for older people, to reduce road congestion and to reduce the negative environmental impact of road traffic. The benefits are to individuals and to the community as a whole.

PRICE CONTROLS

The equilibrium price where demand and supply are equal may not be desirable from the society's point of view. The government therefore intervenes and fixes the price above or below the market equilibrium price.

Price mechanism and price control are two opposite concepts. Government intervening in the price mechanism can distort the situation; so an alternative to

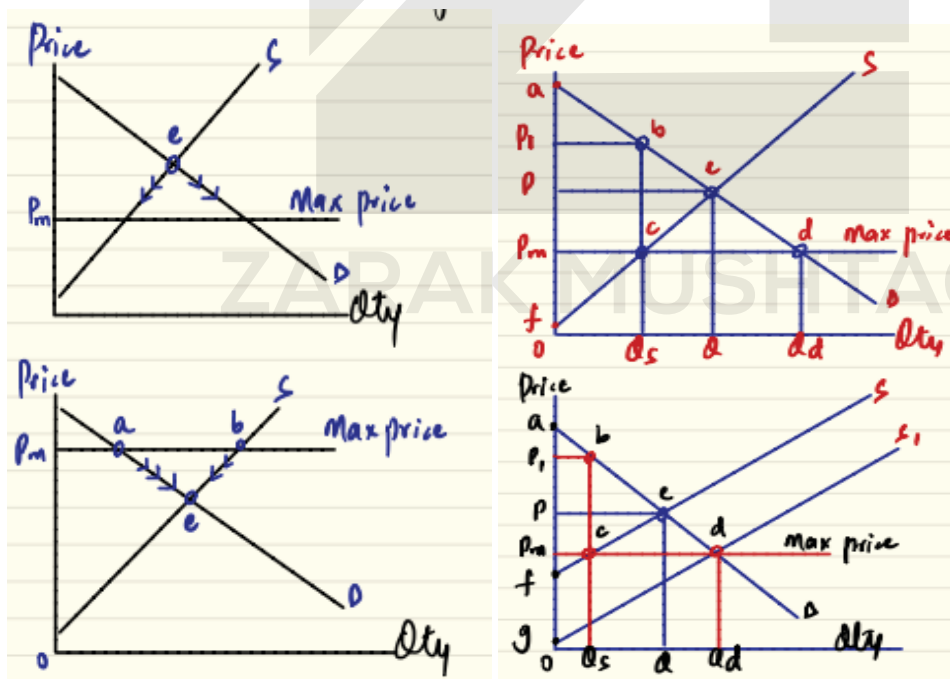
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price mechanism to allocate a good between consumers is price control (e.g. maximum price imposed below the equilibrium)

Maximum price: The government can set a legally imposed maximum price in a market that suppliers cannot exceed - in an attempt to prevent the market price from rising above a certain level. To be effective a maximum price has to be set below the free market price. A price ceiling allows consumers to obtain some essential good or service that they could afford at the equilibrium price because it was too high before. One example of a maximum price might be when shortage/volume of imports of essential foodstuffs threatens large rise in the free market price. Other examples include rent control on properties. Target is to benefit consumers and lower the rate of inflation.

If the government merely sets the maximum price and let the distribution be decided by the private sector without any official intervention, it will lead to certain problems. Goods and services may be allocated on the basis of first come first served which will lead to long queues or long waiting lists. Firms may also discriminate among the customers i.e. they may prefer friends and relatives or regular customers in the distribution of goods and services.



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Black market is the major problem with maximum prices where consumers pay a higher price for the goods which they were unable to get enough in the regular market; organization's revenue will increase.

Maximum price mostly used to solve the basic economic problem and control the level of inflation in the economy. It creates shortage and then black marketing, government need to intervene again by providing subsidies to cover the problems; though maximum price has the potential to increase consumer surplus.

Maximum and Minimum prices

Governments use legislation to enforce maximum prices for:

- Staple food items such as rice and cooking oil
- Petrol and diesel fuel
- Rents in certain types of housing
- Services provided by utilities, such as water, gas and electricity
- Transport fares, especially where a subsidy is being paid.

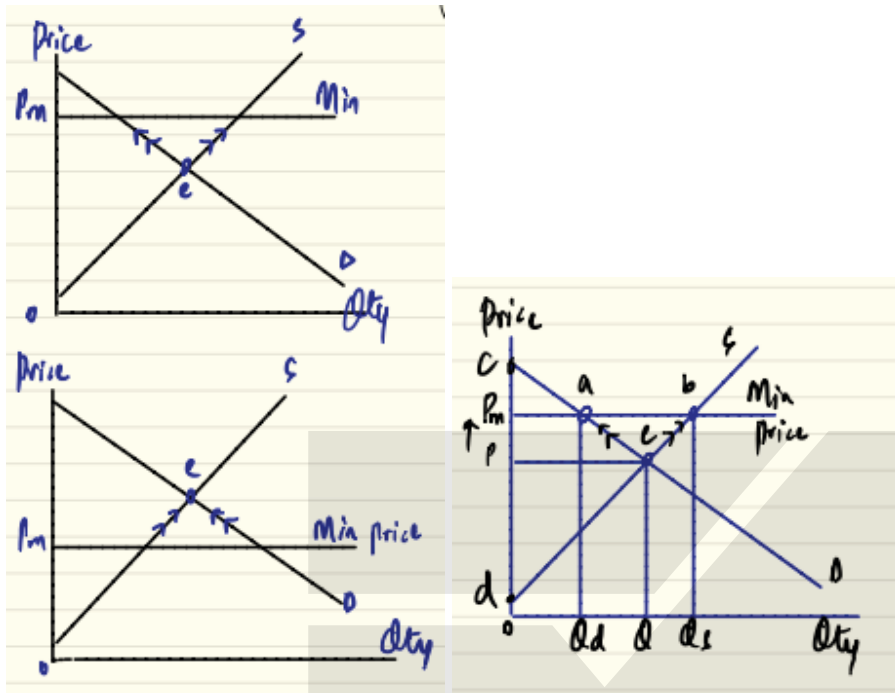
Governments use legislation to enforce minimum prices for:

- Demerit goods such as high-sugar sports drinks
- Agricultural products, particularly in middle and in high-income countries
- Wages in certain occupations, usually low skilled, to avoid employers exploiting their employees
- Certain types of imported goods where close substitutes produced by domestic producers are available.

Minimum price: A minimum price is legally imposed price floor where the normal market price cannot fall. To be effective the minimum price has to be set above the normal equilibrium price. At that price suppliers would be willing to supply more quantity and the demanders will be willing to buy lesser quantity causing a surplus of the commodity. Target is to protect the income of the producers. If minimum price is set below the equilibrium, it will reach back to equilibrium as it will be ineffective.

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Minimum price leads to surplus and government purchase that surplus to maintain the income of farmers but if government did not purchase the surplus; then the farmers' income would be uncertain.

Governments set price floors to protect the welfare of certain suppliers by maintaining a higher price and thus maintain a sufficient income. In addition to agricultural price supports, another form of price floor is minimum wage regulation. The government can adopt many measures to deal with surplus associated with minimum prices.

1. They could restrict supply or increased demand, thereby closing the gap of surplus.
2. They could purchase all the surplus output at market price.

Buffer stock schemes

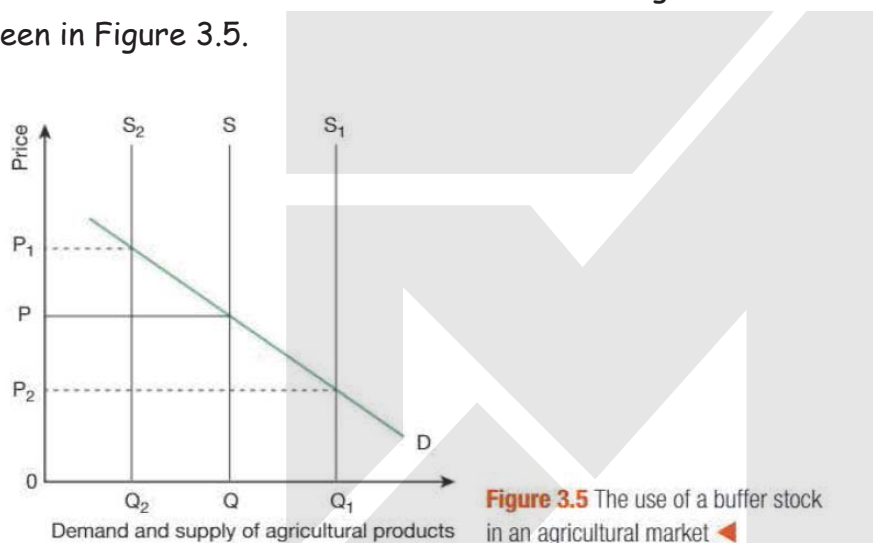
Price stabilization refers to a situation where action needs to be taken' by a government where there are wide fluctuations in price, largely as a result of unplanned fluctuations in supply. This is a particular feature of agricultural

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markets where supply is affected by a variety of factors beyond control, such as bad weather, natural disasters and the effects of pests.

One way that a government could intervene in such a situation is through the establishment of a buffer stock. A buffer stock is where a reserve of a commodity is kept in order to stabilize prices in a market, so that the prices fluctuate within a given price range. When a surplus of a commodity is produced, the product is bought, adding to the buffer stock. When there is a shortage of the product, stocks can be run down. The effect of creating a buffer stock in a market can be seen in Figure 3.5.



The supply of an agricultural product in any one year will be perfectly inelastic. The supply in three years is shown by the supply curves s , s_1 and s_2 . The government, in operating a buffer stock scheme, will want to maintain a price of P and a quantity of Q in the market.

In the year shown by the supply curve s_1 , the equilibrium quantity would have been Q_1 and the equilibrium price would have been P_2 . In this situation, the government would buy the extra output shown by the horizontal distance between Q and Q_1 , and store it. This would keep the price at P .

In the year shown by the supply curve s_2 , the equilibrium quantity would have been Q_2 and the equilibrium price would have been P_1 . In this situation, the government would sell the extra output shown by the horizontal distance between Q and Q_2 , and this would keep the price at P .

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Advantages of buffer stock schemes Disadvantages of buffer stock schemes

- It overcomes the problem of wide fluctuations in prices from one year to another.
- If prices can be stabilized through a buffer stock scheme, there will also be more stability in the incomes that producers, such as farmers, receive.
- This stability in incomes will encourage producers to make long-term plans.
- It may not be easy for the government to establish what the equilibrium price in the commodity market should be. The producers will want the highest possible price to be maintained in the market whereas the consumers will want the lowest possible price to be maintained.
- The cost of operating the buffer stock scheme will need to be paid by somebody. There may be different views about this. For example, the government may decide to pay for the scheme or it may want the producers to contribute to the cost.
- There may be problems if there is a succession of good harvests. For example, it may be difficult to store it all and there may also be concerns about the perishability of the stock. On the other hand, if there is a succession of poor harvests, it may not be possible to store enough supplies.

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